MOTOR CONTROL SYSTEM

Quick Reference Manual

DR28T2.5B03-AZAKD × AZD-KEP Driver

Revision	Platform	Interface	Date
1.0	Ubuntu + ROS2	USB / EtherNet/IP	April 2026

This manual covers hardware specifications, communication methods, ROS2 node parameters, topic API, and the GUI for the Oriental Motor DR28T compact electric cylinder paired with the AZD-KEP EtherNet/IP driver.

1 Hardware Specifications

1.1 Motor — DR28T2.5B03-AZAKD

Parameter	Value	Notes
Type	Compact Electric Cylinder (Ball Screw)	
Stroke	30 mm	Soft limit set to 29 mm in software
Lead / Pitch	2.5 mm / rev	
Max Speed	100 mm/s	Hard cap in motor_controller_node.py
Max Thrust	20 N	
Repeat Accuracy	±0.003 mm	Ball screw shaft tip
Min Travel	0.001 mm	
Encoder	Absolute (Mechanical Multi-turn)	No battery required
Supply Voltage	24 / 48 VDC	Driver dependent

1.2 Driver — AZD-KEP

Parameter	Value
Series	AZ Series Closed Loop Stepper
Fieldbus	EtherNet/IP™ (CT16 conformant)
Network Speed	10 / 100 Mbps (autonegotiation)
Control Power	24 VDC ±5%, 0.25 A
Pulse Input	2 pts (Photocoupler, max 1 MHz)
Control I/O	6 inputs / 6 outputs (Photocoupler)
USB Port	Mini-B → MEXE02 configuration (Windows only)
EDS File	Provided by Oriental Motor

Note: The AZD-KEP's USB port exposes a Virtual COM Port (Windows: COMx, Linux: /dev/ttyACM0). On Ubuntu/ROS2 this is used for MEXE02 frame replay since MEXE02 software is Windows-only.

2 Communication Interfaces

2.1 USB Serial (Current Method — Ubuntu/ROS2)

The motor is controlled by replaying binary MEXE02 protocol frames originally captured on Windows. Frames are extracted from HTML log files generated by MEXE02 and sent over the USB Virtual COM Port.

Setting	Value
Device	/dev/ttyACM0

Baud Rate	19200
Data Bits	8
Parity	Even
Stop Bits	1
Timeout	0.5 s

Position & Speed Encoding

All values are transmitted in **micrometres (µm)**, little-endian 32-bit integers:

Value	Formula	Baseline Hex (little-endian)	Example
Position	mm × 1000	a8 61 00 00 (= 25 mm)	15 mm → 3A 98 00 00
Speed	mm/s × 1000	e8 03 00 00 (= 1 mm/s)	3 mm/s → B8 0B 00 00

XOR Checksum: mask = XOR(baseline bytes) ⊕ XOR(target bytes). New checksums = old checksums ⊕ mask. Implemented in create_combined_html() in motor_controller_node.py.

HTML Template Files Required

File	Purpose
write25mm.html	Baseline position command (25 mm)
1mms.html	Baseline speed command (1 mm/s)
2home.html	Homing sequence frames
start_run.html	Execute motion trigger

2.2 EtherNet/IP (Recommended for Production)

The AZD-KEP's primary interface is EtherNet/IP (CIP). Implicit messaging exchanges 40-byte command and 56-byte status assemblies every 10 ms over UDP.

Direction	Instance	Size	Key Contents
Scanner → Driver (command)	0x65 (101)	40 bytes	Control word, target position (µm), target speed (µm/s)
Driver → Scanner (status)	0x64 (100)	56 bytes	Status word, current position (µm), alarm code

Setup: assign IP via driver rotary switches → connect Ethernet → ForwardOpen with RPI = 10,000 µs, Point-to-Point, Header32Bit on O→T. Python libraries: cpppo or pycomm3 CIPDriver.

3 ROS2 Node — motor_controller_node.py

3.1 Topics

Topic	Type	Direction	Description
/motor_cmd	std_msgs/Float32MultiArray	Subscribe	data[0] = position (mm), data[1] = speed (mm/s)
/motor_home	std_msgs/Empty	Subscribe	Trigger homing sequence
/motor_status	std_msgs/String	Publish	idle | busy | running | error

3.2 Parameters

Parameter	Default	Description
com_port	/dev/ttyACM0	Serial device path
baudrate	19200	Serial baud rate
serial_timeout	0.5	Read timeout (s)
serial_write_timeout	0.5	Write timeout (s)
max_position_mm	29.0	Position upper limit (mm). Physical stroke = 30 mm
max_speed_mms	100.0	Speed upper limit (mm/s). Datasheet rated max
frame_delay	0.1	Inter-frame delay (s)
heartbeat_interval	0.2	Watchdog ping period (s)

3.3 Motor Status Values

Status	Meaning
idle	Node started or shut down. No active command.
busy	Transmitting serial frames to the motor.
running	Heartbeat active — motor is executing motion.
error	Missing template file or invalid command received.

3.4 Quick Commands

Run node (default port)

```
python motor_controller_node.py
```

Run with custom port

```
ros2 run motor_controller_node --ros-args -p com_port:=/dev/ttyUSB0
```

Send motion command

```
ros2 topic pub /motor_cmd std_msgs/msg/Float32MultiArray "{data: [15.0, 3.0]}" --once
```

Trigger homing

```
ros2 topic pub /motor_home std_msgs/msg/Empty "{}" --once
```

Monitor status

```
ros2 topic echo /motor_status
```

4 GUI — motor_gui.py

Run from the motion_cmd directory:

```
python motor_gui.py
```

Element	Function
Status dot + label	Live colour-coded /motor_status feed
Position entry	Input 0 – 29 mm. Validated before publish.
Speed entry	Input 0.1 – 100.0 mm/s. Validated before publish.
Send Command button	Publishes position + speed to /motor_cmd
Go Home button	Confirmation dialog → publishes to /motor_home
Log box	Shows all outgoing publishes and incoming status

Status indicator colours:

Colour	Status	Meaning
● Green	idle	Motor ready, no active motion
● Orange	busy	Frames being sent over serial
● Blue	running	Heartbeat active, motor moving
● Red	error	Command rejected or file missing

Requires: rclpy, tkinter, std_msgs. ROS2 spin runs in a daemon thread — closing the window shuts down the node cleanly.

5 Safety & Operating Limits

Limit	Value	Enforced By
Max position	29.0 mm	motor_controller_node (parameter) + GUI validation
Min position	0.0 mm	motor_controller_node + GUI
Max speed	100.0 mm/s	motor_controller_node (parameter) + GUI validation
Min speed	> 0.1 mm/s	motor_controller_node + GUI
Physical stroke	30 mm	Hardware — do not override max_position_mm above 29

Important Warnings

▲ Always home the motor after power-on before sending position commands.

▲ Do not set max_position_mm above 29.0 mm — the physical stroke is 30 mm and over-travel will damage the cylinder.

▲ Ensure /dev/ttyACM0 permissions are correct on Ubuntu: sudo chmod 666 /dev/ttyACM0

▲ The heartbeat (watchdog) must be running at all times. If the node crashes, the motor will stop receiving pings and may fault.

▲ Only one process should hold /dev/ttyACM0 at a time. Close MEXE02 (Windows) before running the ROS2 node.

6 Troubleshooting

Symptom	Likely Cause	Fix
ModuleNotFoundError: rclpy	ROS2 environment not sourced	source /opt/ros/humble/setup.bash
Failed to open /dev/ttyACM0	Permission denied or cable unplugged	sudo chmod 666 /dev/ttyACM0
No response after sending frames	Wrong baud / parity or motor not powered	Check 24 V supply; verify serial settings
Position rejected (out of range)	Value exceeds max_position_mm (29.0)	Send value between 0.0 and 29.0 mm
Speed rejected	Value exceeds 100.0 mm/s or ≤ 0	Send value between 0.1 and 100.0 mm/s
Motor stops mid-motion	Heartbeat thread died (SerialException)	Restart node; check USB connection
GUI shows no status updates	/motor_status not published	Confirm motor_controller_node is running
FileNotFoundError: write25mm.html	Node launched from wrong directory	cd motion_cmd before running node